THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM THREE ANNUAL EXAMINATION
041 BASIC MATHEMATICS

Time: 3 Hours **November 2025**

Instructions

1. This paper consists of sections A and section B with a total of 14 questions.
2. Attempt **all** questions in both sections.
3. Each question in section A carries 6 marks while each question in section B carries 10 marks.
4. Mathematical tables and non-programmable calculator may be used.
5. Write your examination number on every page of your answer sheets.

SECTION A (60 MARKS)

1. (a) Convert $0.\overline{135}$ into the simplest fraction.
   (b) Subtract 5% of 300 000 shillings from 35% of 600 000 shillings.

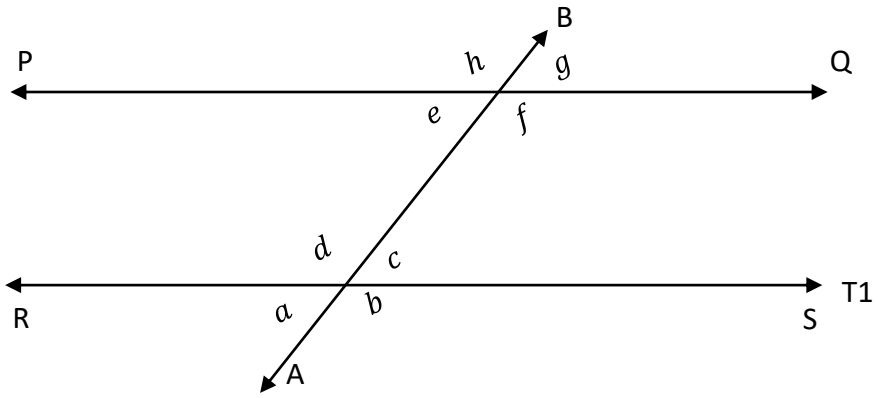
2. (a) Solve the following exponential equation: $\left(\frac{3}{2}\right)^{n-1} = \left(\frac{27}{8}\right)^{n+3}$.
   (b) Evaluate $\log\left(\frac{764 \times 36.56}{\sqrt{3845}}\right)$, given that $\log 7.64 = 0.8831$, $\log 3.656 = 0.5630$, and $\log 3.845 = 0.5849$.

3. (a)Every woman in Seypalm club owns a land rover or a bicycle. If 23 women own a land rover, 14 own bicycle and 5 own both a land rover and a bicycle, how many women are in the club?
   (b)Find the union and intersection of the following pair of sets:
   $A = \{All\ multiples\ of\ 5\ less\ than\ 30\}, B = \{All\ multiples\ of\ 10\ less\ than\ 30\}$.

4. (a)Find the equation and $y - intercept$ of a line passing through the following pairs of points: $(4,4)\ and (-3,2)$.

(b) From the following figure where by $\overline{PQ}$ is parallel to $\overline{SR}$ and $\overline{AB}$is a transversal.

Figure 1

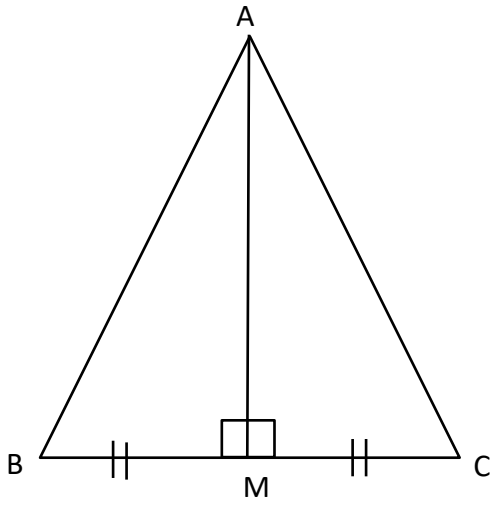


Name angles which are:

i. Corresponding angles
ii. Alternate interior angles
iii. Alternate exterior angles
iv. Vertically opposite angles

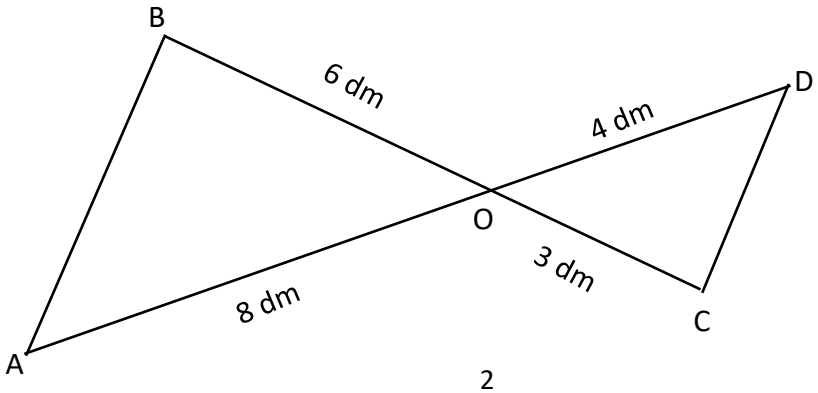
5. (a) Use the following figure to prove that $\angle BMA = \angle CMA = 90^{\circ}$, given that $\overline{AB} = \overline{AC}$

Figure 2



(b) In the following figure, if $\overline{OA} = 8dm, \overline{OB} = 6dm, \overline{OD} = 4dm$ *and* $\overline{OC} = 3\ dm$. Prove that $\Delta AOB \sim \Delta DOC$

Figure 3



6. (a) The following table shows the value of y for some selected values of x. The variables $x\ and\ y$ are connected by the relation $''y$ varies inversely asx''. Calculate the missing values of$y$.

x	5	10	15	20
y				1.5

(b) A bus travels 480 km using 32 litres of diesel. How many kilometers will be covered by 20 litres of diesel? (Other things are constant).

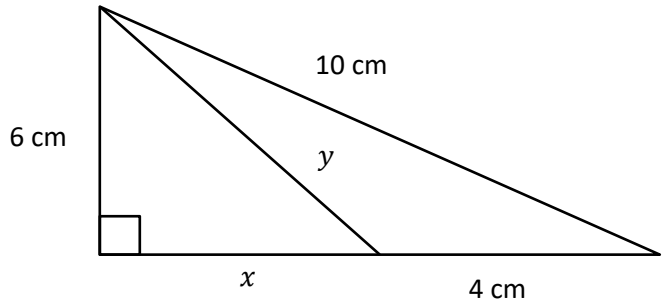
7. (a) When 3 million shillings is shared into the ratio of 2: 3: 5, what is the share of the second proportional?
(b) Record the following transaction in the cash book and balance it

January 1 started business with capital in cash sh. 500 000/=
January 2 bought office machines for cash sh. 150 000/=
January 5 purchased goods for cash sh. 2000 000/=
January 18 received a cash loan from Seypalm of sh. 3,000 000/=
January 23 sold good for cash for sh. 120 000/=
January 29 sold good to mary 280 000/=

8. (a) Insert three arithmetic means between 18 and 30
(b) Three numbers: $x-2, x, and\ x+3$ are three consecutive terms of a geometric progression. Find the sum of the first six terms.

9. (a) Express $\tan 105^\circ = \tan(60^\circ + 45^\circ) = \frac{\tan 60^\circ + \tan 45^\circ}{1 - \tan 60^\circ \tan 45^\circ}\ in\ form\ of - a\left(b + \sqrt{c}\right)$

(b)Find the values of $x\ and\ y$ in the following figure:



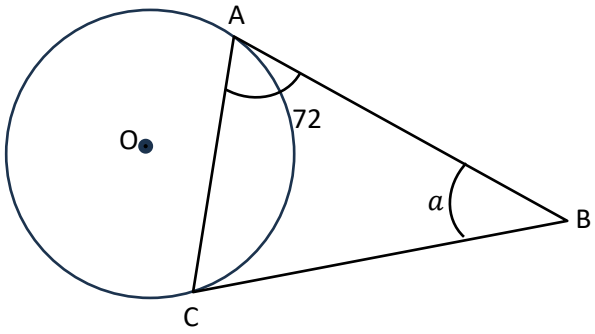
10. (a) Graph the solution of $2(2x+3) - 10 < 6(x-2)$.
(b) Find two consecutive odd numbers whose product is 195.

SECTION B (40 MARKS)

11. (a) The mass of 42 athletes were recorded as shown in the following table. Use it to calculate the measure of the central tendency (mean, median, and mode). Leave the answer to 2 decimal places

Mass in kg	$40-44$	$45-49$	$50-54$	$55-59$	$60-64$	$65-69$
frequency	7	8	11	10	4	2

(b) Calculate $angle\ a$ in the following figure where by $\overline{AB}\ and \overline{BC}$ are tangents to circle O



12. (a) Two towns P and Q are 2 500 km apart, point P being due south of point Q. If the latitude of P is $5^{\circ}S$, find the latitude of point Q.
    (b) Two towns A and B are located at $(20^{\circ}S, 38^{\circ}E)$ and $20^{\circ}S, 43^{\circ}E$ respectively. Find the distance between the two towns in kilometers. Use
    $1\ nautical\ mile = 1.852\ km\ and\ leave\ the\ answer\ to\ 2\ decimal\ places.$

13. (a) A translation takes every point a distance of 1 unit to the left and 2 units downward on the $xy-$ plane. Find where it takes$(3\ ,7)$.
    (b) Find the image of point Q (5,2) after a reflection in the line$y + x = 0$.

14. (a) Given $k(x) = \begin{cases} -3\ if\ x \leq 1 \\ 1\ if\ 1 < x \leq 2 \\ 4\ if\ 2 < x \end{cases}$

    i. Sketch the graph of $k(x)$.
    ii. State the domain and range of $k(x)$.
    iii. Find the value of $k(-1)$.

    (b) Given $f(x) = x^2 - 2x - 3$
    i. State the domain and range of f
    ii. Equation of symmetry
    iv. Minimum or maximum value of f